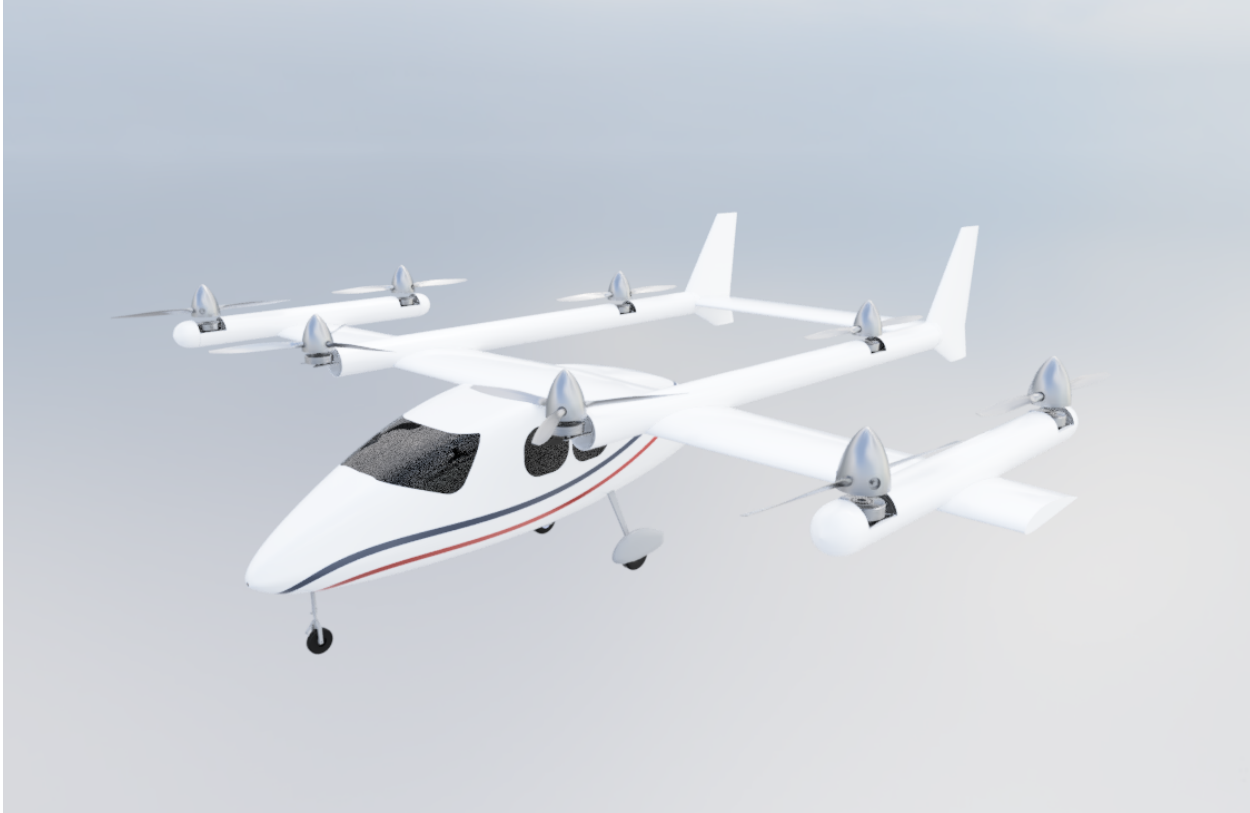
Report III

Conceptual design



April 5, 2024

Contents

1 Matching chart 6
1.1 Requirements revision 6
1.2 Results 6
1.3 Design space and comparison 8
2 Aerodynamic estimation 10
2.1 Wing geometry 10
2.2 Airfoil selection 11
2.3 Tail sizing 12
3 Configuration layout and lofting 14
3.1 Fuselage 14
3.2 Cabin interior 14
4 Structural and material considerations 16
4.1 Monocoque or semi-monocoque structure 16
5 Propulsion system selection 18
5.1 Preliminary selection: Emrax 348 19
6 Propeller design 21
7 Energy system 23
7.1 Requirements 23
7.2 Available in the market 23
7.3 Proposed arrangement 24
8 Weight engineering 26

9 Stability and control estimation analysis 27
10 Conclusion 33

List of Figures

1.1 Wing loading vs Thrust-To-Weight diagram 7
1.2 Power vs Wing loading 8
2.1 General characteristics of the wing 11
2.2 Wing loads distribution 11
2.3 Airfoils anlysis 12
3.1 Fuselage layout and loft 14
3.2 Layout of passengers 14
3.3 Layout of seats up view 14
3.4 Cockpit general dimensions 15
4.1 Fuselage structure 17
5.1 Emrax 348 motor. Taken from [3] 19
5.2 Power and Torque vs. RPMs for Emrax 348 20
5.3 Efficiency map for Emrax 348 20
6.1 2 and 3 blades distribution 22
6.2 3 blades propeller 22
6.3 2 blades propeller 22
7.1 Arrangement of cells 24
7.2 Battery package 25
9.1 Static stability 28
9.2 Lateral modes plot 29
9.3 Longitudinal modes plot 30

9.4 modes 30
9.5 thumbprint 32

List of Tables

1.1 Similar aicraft comparison 9
2.1 Tail sizing 13
4.1 Comparison between Monocoque and Semi-Monocoque Structures 16
5.1 List of analyzed motors and their characteristics 18
6.1 Propeller Design Parameters 21
7.1 Battery Requirements 23
7.2 Battery Specifications by Supplier 23
8.1 Weights 26

1 Matching chart

This section will use a tool that allows to determine how different performance parameters delimit a feasible design region, allowing different options of optimal configurations that should be guiding at design time rather than restrictive. This method is called constraint analysis [1].

1.1 Requirements revision

In order to perform this analysis, parameters provided by the client and other aircraft references similar to the above mentioned concept were taken into account.

Factors such as the stall speed of 31 m/s was defined in advance. Other factors such as nominal bank angles of 60° while maintaining a cruise speed of 70 m/s, a takeoff run of 285 m considered STOL by EASA and an operating altitude of 1000 ft above the ground. A maximum climb angle of 10° was also considered since, being an aircraft with fast take-off capability, it must gain its operating altitude quickly and avoid any obstacle on the ground. When the aircraft is in VTOL operation, this design feature will not significantly interfere with its design. With these desired parameters, we continue using the equations found and developed in section 3 of [1].

1.2 Results

These are the results obtained from the analysis performed:

In the graph above we see how the performance constraints determine a feasible design region, in this case, the white region. This region is all the design possibilities, however, if an optimum point is desired, the lowest Thrust-to-weight ratio should be sought, since this is usually related to lower costs and higher payload capacity. In addition, if we want to have a wide range of MTOW we should aim at the highest wing loading point (this will have implications in its maneuverability and flight characteristics), beyond this point it is not an effective design if it is not compensated with thrust. In addition, a historical review

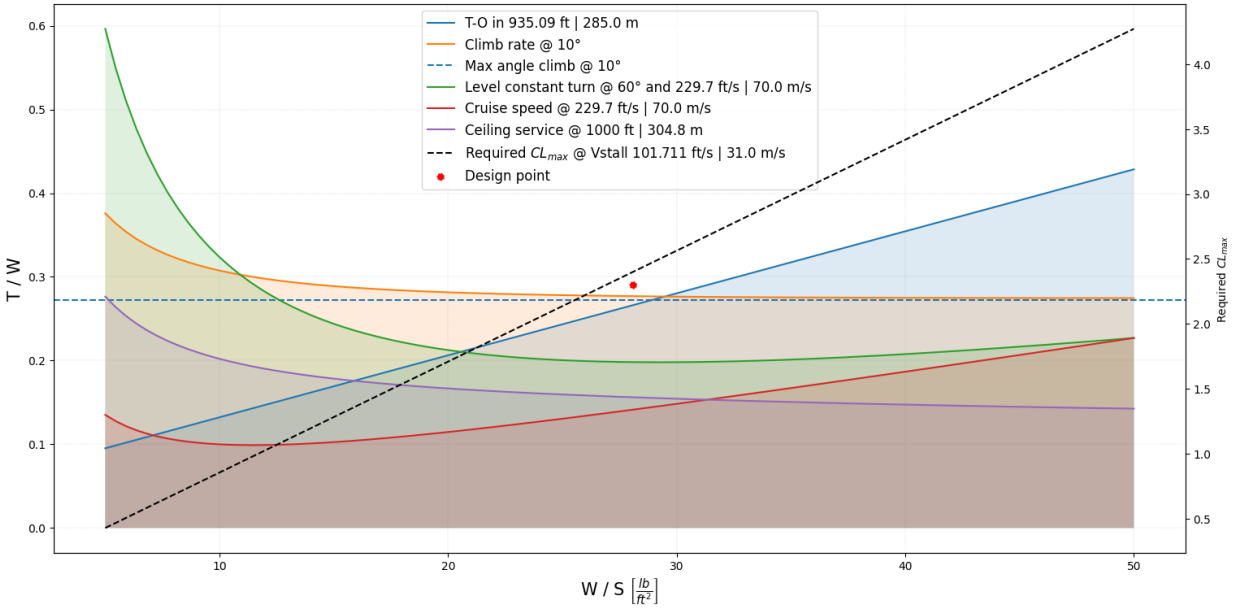


Figure 1.1: Wing loading vs Thrust-To-Weight diagram

of general aviation aircraft lift coefficients determined that values greater than 2.4 are not easily achievable and increase the complexity of the design. This establishes a new constraint in the design region.

Thus, a design point with a Wing loading (W/S) of $29.57 lb/ft^2$ was chosen, which has a thrust-to-weight ratio of 0.29, reasonable and common values in general aviation aircraft.

Using the above results, we proceed to transform the thrust-to-weight ratio using the initial estimate of $\sim 2900 kg$ to power number in horsepower. This for each flight stage, i.e., using their velocities, density and altitude changes. Obtaining the following:

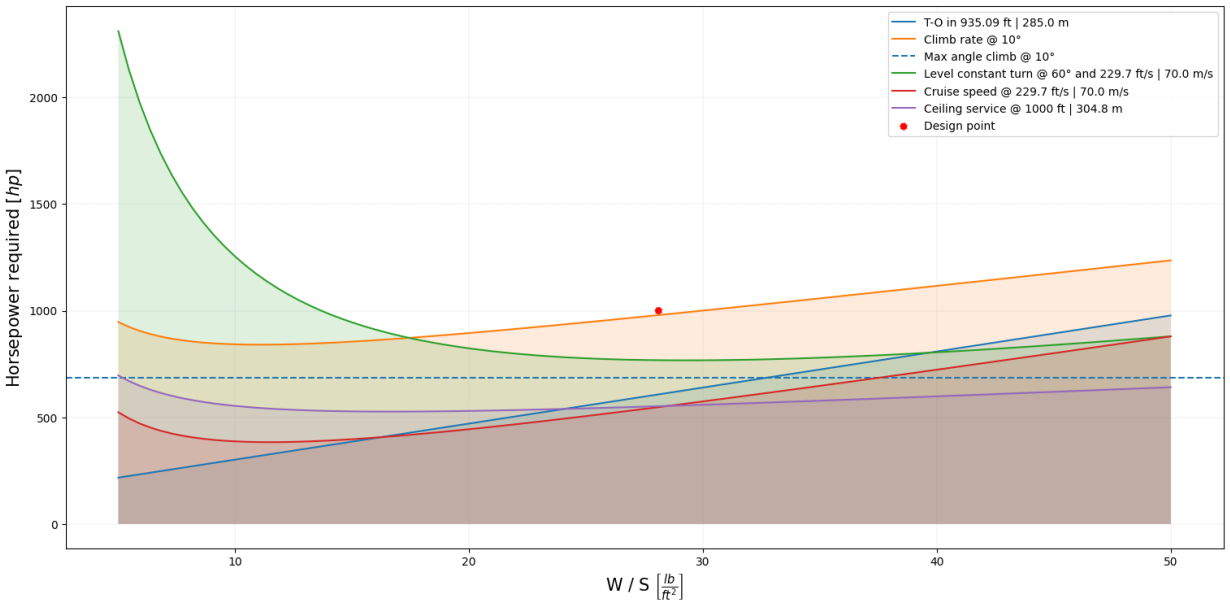


Figure 1.2: Power vs Wing loading

Thus we find that for our design point a power close to 1000 hp is required. This estimate will later be verified by another method.

1.3 Design space and comparison

To verify our results, we compared them with aircraft already in service such as the Cessna Caravan, Cessna Skymaster and other aircraft shown below. This indicates that the capabilities of the aircraft will be similar to these others and we can trust the selection of this design point.

These data were taken using each aircraft's MTOW, wing area and propulsion system power. Also, similar aircrafts can be found in [2].

Aircraft	**W/S** $[lb/ft^2]$	**Power** $[hp]$
Cessna Caravan	31.4	675
Cessna Skymaster	21.9	420
BN-2 Islander	24.2	800
IAI Arava	32	1400
Dornier Do 28	28.4	760
DHC-6 Twin Otter	29.8	1156
Design point	29.6	990

Table 1.1: Similar aicraft comparison

2 Aerodynamic estimation

In this section, the sizing of the lifting surfaces that govern the behavior and performance of the aircraft, the main wing and the stabilizers is carried out. For this purpose, the constraint analysis previously performed is used, implementing the estimated weight.

$$W/S = 29.6\frac{lb}{ft^2}$$

$$\frac{6393.4lb}{29.6\frac{lb}{ft^2}} = S$$

$$S \approx 216ft^2 \approx 20m^2$$

This value is very close to the value estimated in the previous conceptual design report, what is done next is to modify the wing geometry to take full advantage of its capabilities.

2.1 Wing geometry

For the definition of the wing geometry, an aspect ratio of 8 was determined, a very common value in general aviation aircraft. Some values are usually higher than this, however, given the characteristic of having engines dedicated to VTOL it is not convenient to have a very high wingspan due to the moments and loads generated by the thrust and weight of these engines, in addition to the requirement for vertiports with small space available.

To comply with the area found S, the chords are varied along the span of the wing, looking for good performance characteristics and stall prevention or at least, a controlled stall using tools such as dihedral and twist both aerodynamic and geometric. This procedure was made using XFLR5.

As we can see, the leading edge is maintained to facilitate the manufacturing process, however, for the trailing edge, varieties of chord lenghts are used to achieve an elliptical lift distribution, meaning, a more efficient design and utilization of its area.

This solution makes it possible to obtain the best possible distribution without the complexity of manufacturing an elliptical wing.

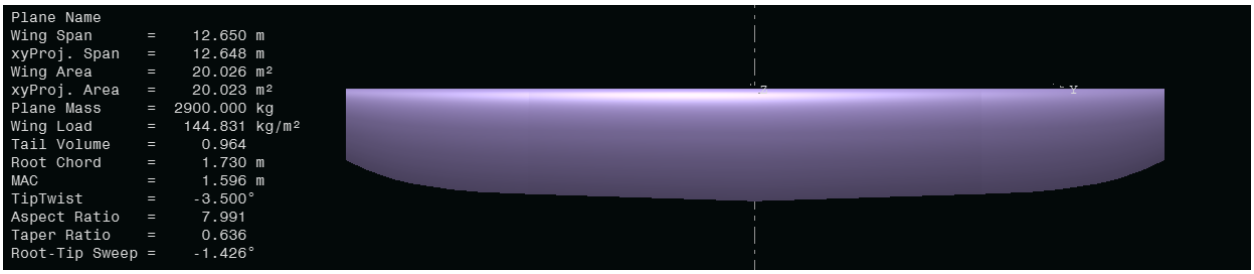


Figure 2.1: General characteristics of the wing

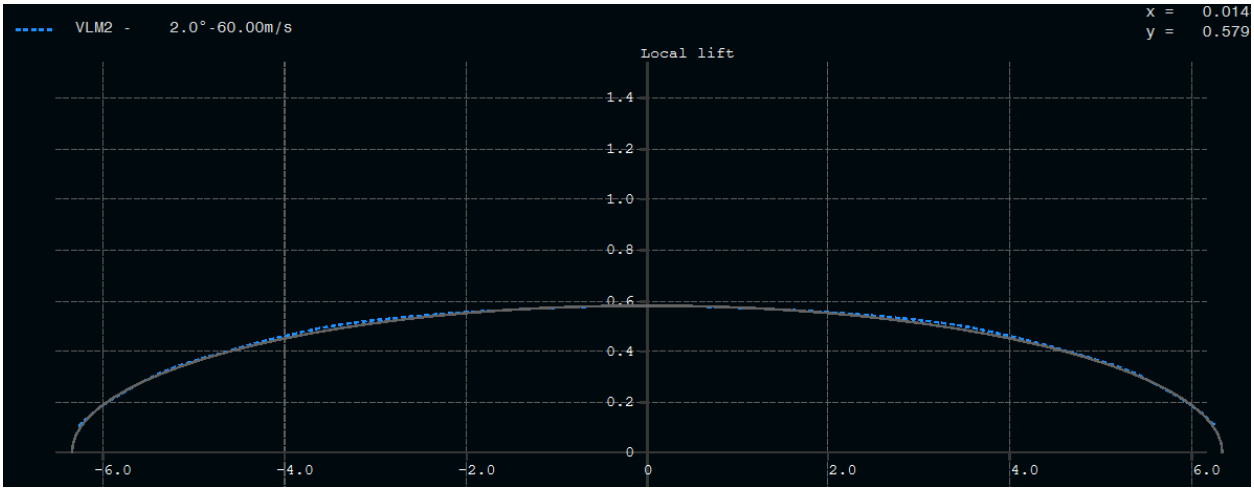


Figure 2.2: Wing loads distribution

2.2 Airfoil selection

Given the required performance parameters we can find the target lift coefficients for both take-off and cruise conditions. In this case, as can be visualized in 1.1 in order to take in that desired distance the black dashed line indicates a 2.4 $C_{L,max}$. Now, for a desired cruise speed can be derived from:

$$L = \frac{1}{2}\rho C_l S v^2$$

$$C_{l,cruise} = \frac{2 * W}{\rho S v^2}$$

$$C_{l,cruise} \approx 0.5$$

In order to find a proper airfoil that suits our requirement, various airfoils of general aviation aircrafts were analyzed using XFLR5 at a wide range of number of Reynolds.

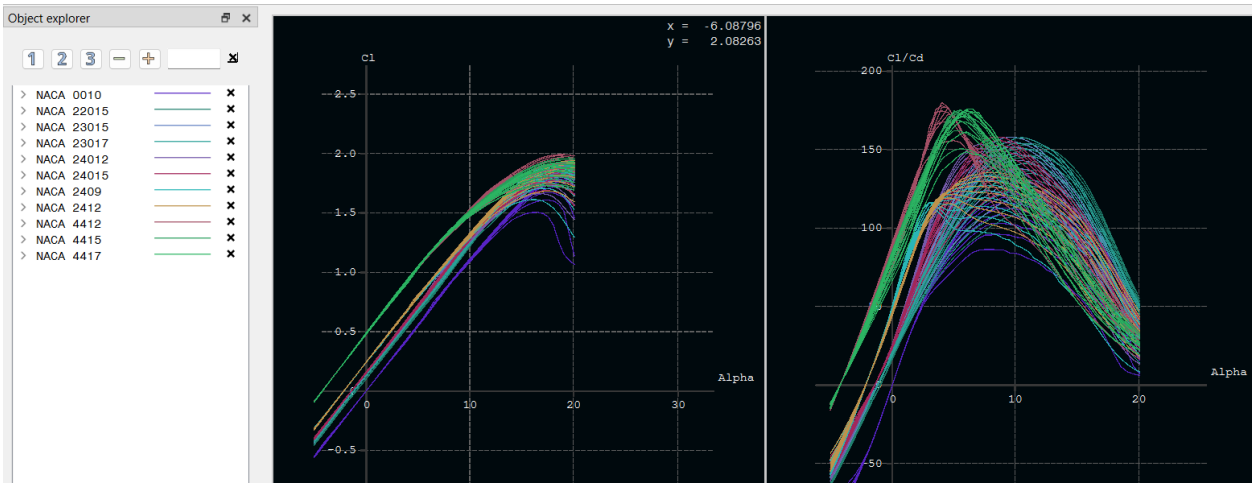


Figure 2.3: Airfoils anlysis

This suggest that the NACA 4417 could be used in our design. Also some variations in thicknes are used in order to polish the lift distribution (NAcA 4415 and 4412). For the stabilizers a symetric airfoil was used, NACA 0010. Finally, geometric twist (washout) were also used to prevent lost of control in ailerons due excesive angles of attack.

2.3 Tail sizing

For the sizing of the stabilizing surfaces, chapter 11 of Professor Gudmundsson's book [1] was used as a guide, which helps to perform an initial sizing of the horizontal and vertical stabilizer. However, the span of the horizontal stabilizer was fixed given the arrangement of the tail booms, restricting this parameter, the chord must be found to give us the necessary area at first sight, since this will be verified with a stability analysis against certain disturbances.

The results of these analyses were:

Parameter	**Vertical stabilizer**	**Horizontal stabilizer**
Volume	0.06	0.98
Taper ratio	0.4	1
Aspect ratio	1.45	4
Tail arm	5.59 m	5.59 m
Surface	$2.71m^2$	$5.54m^2$
Span	1.40 m	4.57 m
Mean geometric chord	0.97 m	1.21 m

Table 2.1: Tail sizing

3 Configuration layout and lofting

3.1 Fuselage

To create the external geometry of the fuselage, various parameters were considered: having curves that made the streamlines as smooth as possible, having a spacious interior without significantly compromising the weight, and thinking in a possible modular interface for future versions. For reference, there is inspiration in the Tecnam 2006T, with modern curve lines and elongated body. Because this is a twin-boom, the fuselage will not joint with the tail, so it is shorter but the curves will not abruptly interrupt.

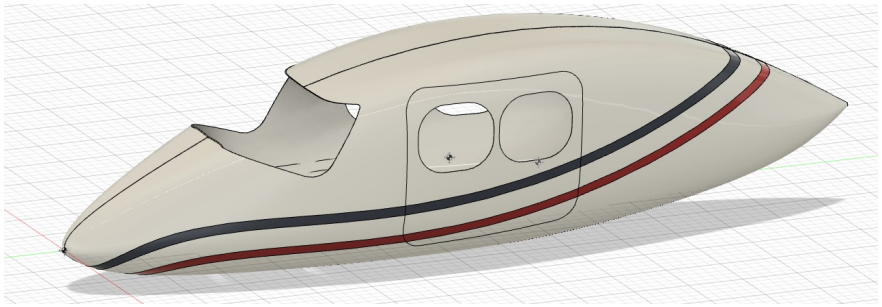


Figure 3.1: Fuselage layout and loft

3.2 Cabin interior

The concept for the cabin is inspired by a business jet comfort, for reference, the interior of a C90 King Air, or Cessna Caravan but with fewer seats.

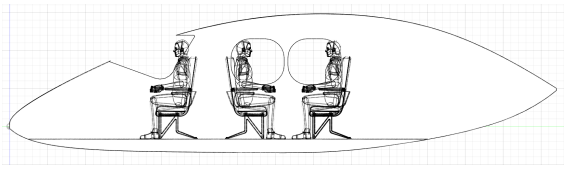


Figure 3.2: Layout of passengers

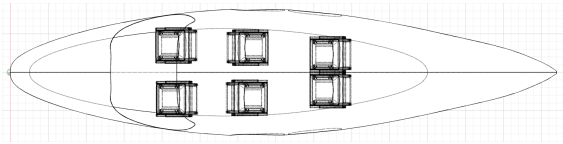


Figure 3.3: Layout of seats up view

For the cockpit, Figure 3.4 shows the general dimensions suggested by the Gudmunssons's book

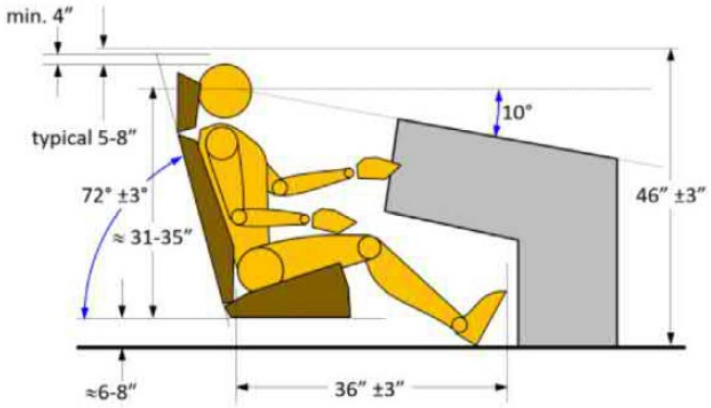


Figure 3.4: Cockpit general dimensions

4 Structural and material considerations

4.1 Monocoque or semi-monocoque structure

For selecting a the best configuration for this application, we construct a table with key characteristics.

Characteristic	**Monocoque**	**Semi-Monocoque**
Structural Integrity	Adequate	Enhanced
Weight Efficiency	High	Adequate
Energy Absorption	Adequate	Improved
Stiffness and Stability	Adequate	Improved
Design Flexibility	Limited	Enhanced
Load Distribution	Concentrated	Distributed
Corrosion Resistance	Material Dependent	Material Dependent
Vibration Damping	Adequate	Enhanced
Maintenance and Repair	More Difficult	Easier
Modularity	Lower	Higher
Complexity of Manufacturing	Lower	Higher
Cost	Generally Lower	Generally Higher

Table 4.1: Comparison between Monocoque and Semi-Monocoque Structures

Based on Table 4.1, we selected the semi-monocoque structure for its superior characteristics in several key areas. This design enhances structural integrity by effectively distributing stresses through the combination of skin and frame support, thereby minimizing the need for heavy internal structures and contributing to weight efficiency. It also excels in energy absorption, providing better protection in crash scenarios through effective distribution of impact forces. Furthermore, the semi-monocoque structure offers higher stiffness and stability, essential for dynamic performance and control. Its design allows for greater flexibility

and modularity, facilitating easier maintenance, upgrades, and adaptation to diverse configurations. This approach simplifies access to internal components for repairs without relying solely on the outer skin, making it a robust and versatile choice for complex applications. Figure 4.1 shows a initial air-frame layout with formers and stringers.

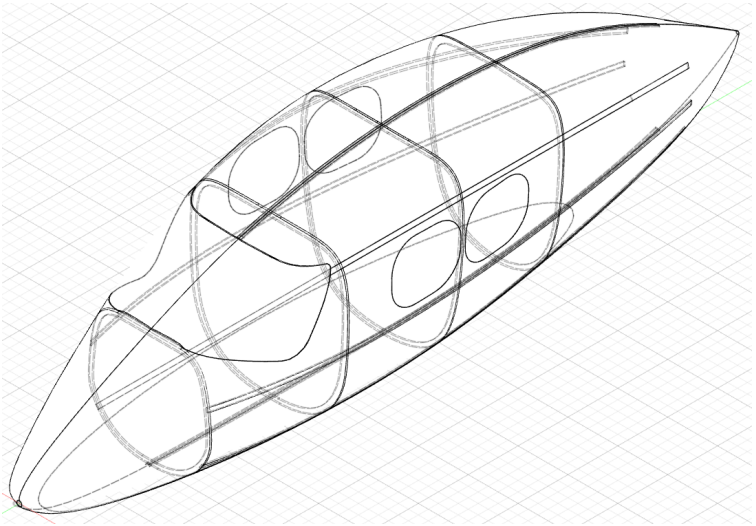


Figure 4.1: Fuselage structure

5 Propulsion system selection

A comprehensive research was performed to select the correct electric motor for the aircraft. Some of the main motor manufacturers and their solutions are listed in Table 5.1. This information was collected from [3], [4], [5], [6], [7].

Table 5.1: List of analyzed motors and their characteristics

Motor name	Continuous Power (kW)	Peak Power (kW)	Mass (kg)	Continuous Torque (Nm)	Peak Torque (Nm)	Power to Weight ratio (kW/kg)	Torque to Weight ratio (Nm/kg)
Emrax 188	37	60	7	56	100	5.285714286	8
Emrax 208	56	86	9.1	90	150	6.153846154	9.89010989
Emrax 228	75	124	12	130	230	6.25	10.83333333
Emrax 268	117	210	20	250	500	5.85	12.5
Emrax 348	210	400	41	500	1000	5.12195122	12.19512195
MAGicALL MAGiDRIVE 6	5	6	0.7	10.8	18	7.142857143	15.42857143
MAGicALL MAGiDRIVE 12	10	12	1.5	28.8	48	6.666666667	19.2
MAGicALL MAGiDRIVE 20	16	20	3	60	100	5.333333333	20
MAGicALL MAGiDRIVE 40	32	40	5	120	200	6.4	24
MAGicALL MAGiDRIVE 75	60	75	9	300	500	6.666666667	33.33333333
MAGicALL MAGiDRIVE 150	120	150	16	720	1200	7.5	45
MAGicALL MAGiDRIVE 300	240	300	30	1440	2400	8	48
MAGicALL MAGiDRIVE 500	400	500	50	3000	5000	8	60
Magnix magny350 EPU	350	-	111.5			3.139013453	0
Magnix magny650 EPU	640		200			3.2	0
Siemens SP90G	65		13			5	0
Siemens SP70D	92		26			3.538461538	0
Siemens SP55D	72		26			2.769230769	0
Siemens SP200D	204	204	49	1500	1500	4.163265306	30.6122449
Siemens SP260D	260		50	1000	1000	5.2	20
Siemens SP260D-A	260		44			5.909090909	0
Yuneec Power Drive 10	10		4.5			2.222222222	0
Yuneec Power Drive 20	20		8.2			2.43902439	0
Yuneec Power Drive 40	40		19			2.105263158	0
Yuneec Power Drive 60	60		30			2	0
MGM-COMPRO 80	65	80	23.8		300	2.731092437	0
MGM-COMPRO 70	45	70	17.3		200	2.601156069	0
MGM-COMPRO 60	40	60	15.3		250	2.614379085	0
MGM-COMPRO 50	35	50	12		200	2.916666667	0

The preselected motors are the ones highlighted in yellow and the selection criteria was:

- Minimum continuous power: 200 kW
- Minimum continuous torque: 500 Nm
- High power and torque density
- High power and torque at low RPMs

5.1 Preliminary selection: Emrax 348

It is a motor produced by Emrax [3] that can be see in Figure 5.1. It was selected for this analysis due to the high availability of information, but other promising companies could provide the aircraft with the solution, such as MGM-COMPRO and magniX. Further work with the selected company is required.

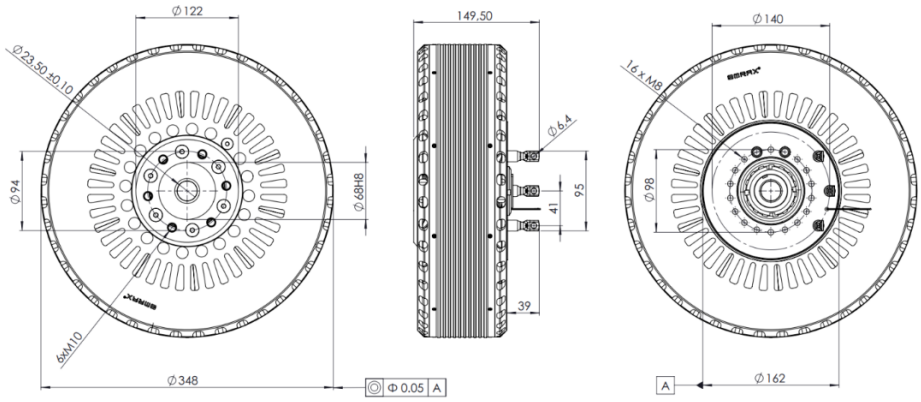


Figure 5.1: Emrax 348 motor. Taken from [3]

The Emrax 348 complies with the initial requirements but will not fulfill the requirement of high power and torque at low rpms as shown in Figures5.2. The benefit of using an axial flow motor is its high efficiency, as shown in Figure 5.3.

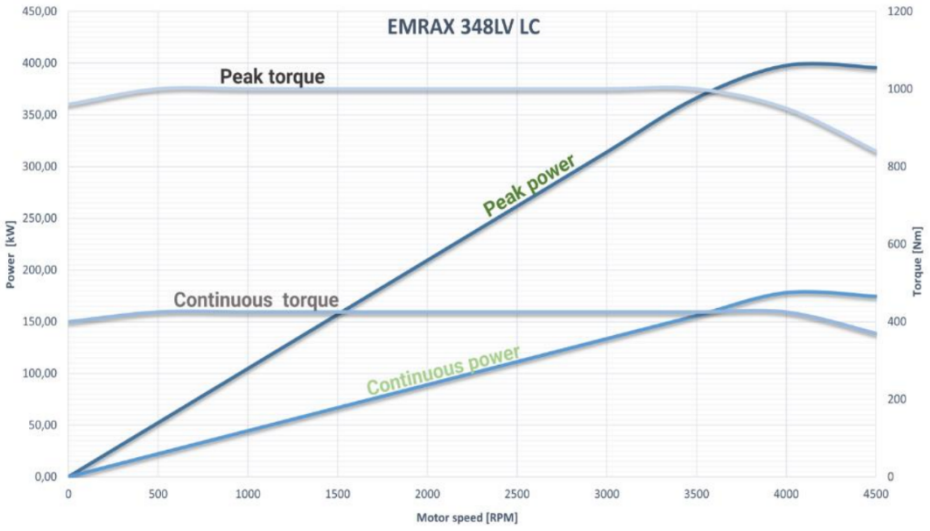


Figure 5.2: Power and Torque vs. RPMs for Emrax 348

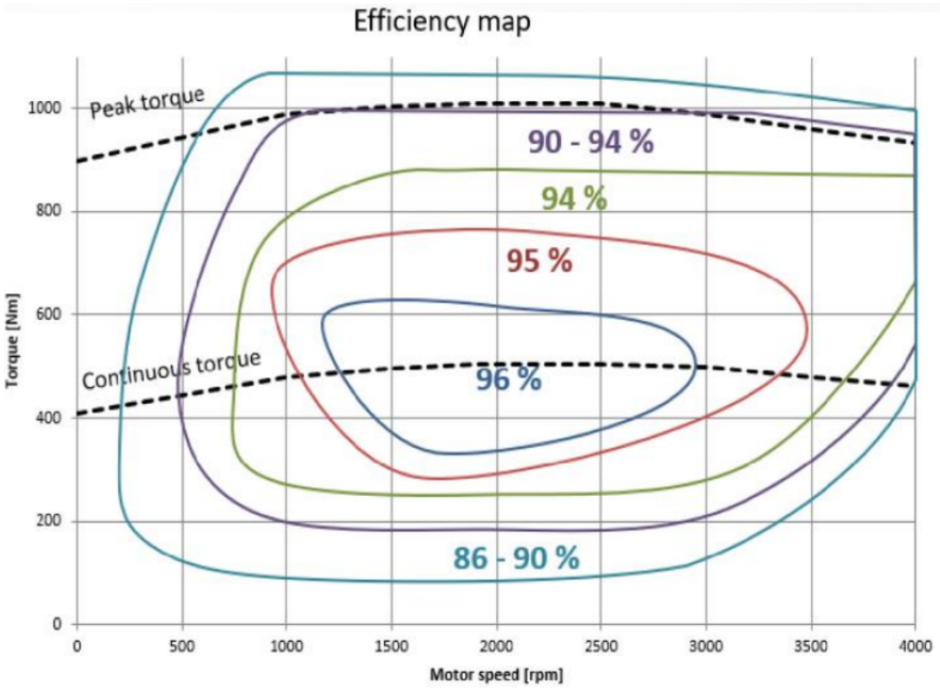


Figure 5.3: Efficiency map for Emrax 348

6 Propeller design

The main requirements for the propeller design are the number of blades and the power they must transmit. With that information, we can calculate its diameter as said by Raymer [8]:

$$D_{prop} = K_p \sqrt[4]{P}$$

Where the Kp is determined by the number of blades: 0,56 for 2 blades and 0,52 for 3.

With the power requirement, the analysis of the diameter and allowable tip speed is performed in Table 6.1. The allowable tip speed was set to 213 due to the low noise requirement for the aircraft. This value is recommended by [8].

Table 6.1: Propeller Design Parameters

Blades	**Power (kW)**	**Kp**	**Diameter (m)**	**Allowable Vtip (m/s)**	**RPM**
2	200	0,56	2,11	213	1932
3	200	0,52	1,96	213	2080
4	200	0,49	1,84	213	2208

Furthermore, the configuration of the aircraft determines that two of the motors will require tilting capabilities (for lift and cruise) and 6 of them will require only lifting capabilities, so the configuration is shown in Figure 6.1. The two inner motors will have 3 blades for more efficiency during cruise and a lower diameter (facilitating the tilting mechanism) and the other 6 motors will have 2 blades to have an aerodynamic positioning during cruise flight.

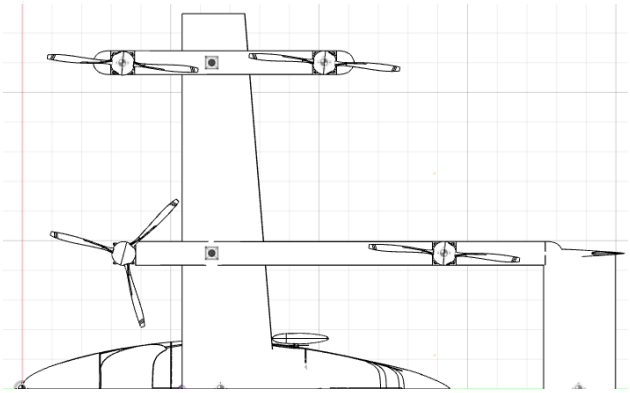


Figure 6.1: 2 and 3 blades distribution

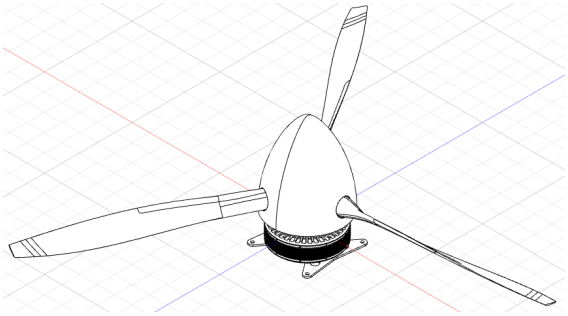


Figure 6.2: 3 blades propeller

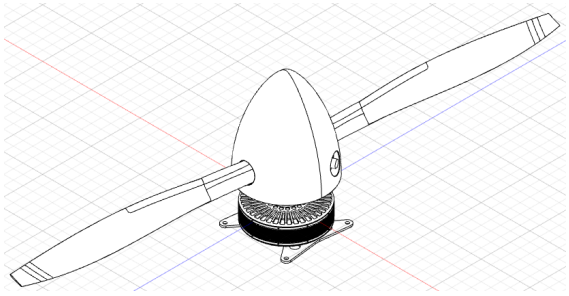


Figure 6.3: 2 blades propeller

7 Energy system

7.1 Requirements

Once the power requirement is calculated from the propulsion system, the following requirements are met:

Parameter	**Value**
Capacity	230 KWh
Minimum density	230-250 Wh/kg
Continuous rate	3C
Pulse Rate	15C

Table 7.1: Battery Requirements

7.2 Available in the market

The battery technology is continuously trying to come up with advancements, and many companies work in hand with the batteries manufacturers to satisfy their specific needs. So, most of the time there is not a product off-shelf that is going to be tailored to the project. In the search for companies, the most promising are in the :

Supplier	**Reference**	**Cell**	**Pack**	**Specific Energy (Wh/kg)**	**Energy density (Wh/L)**
AMPRIUS	SIMAXXTM	✓		320-500	805-1300
MGM COMPRO	LI-ION BATTERY SYSTEM		✓	244	-
MOLICEL	INR21700-P45B	✓		242	643

Table 7.2: Battery Specifications by Supplier

7.3 Proposed arrangement

The drivers to selecting the space for the batteries are: having the batteries as close as possible to the electric motor, having them in an accessible place in case of needing to replace or maintenance, and helping the location of the CG. An initial placement was thought in the booms, making an cylindrical shape with the cells as shown in Figure7.1. This representation was made with the reference INR21700-P45B mencioneted in Table 7.2, since the CAD was available and the supplier gives a energy density in volume.

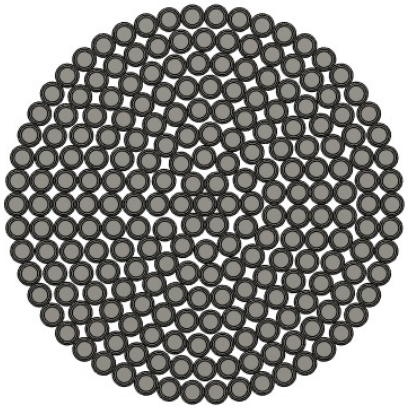


Figure 7.1: Arrangement of cells

Then, with the mentioned distribution 225 cells could be fit in one cross- section. In total we would need a 14375 cells which means 64 layers of the same cross-section, and diving them in 8 packages, each one would be of 1800 cells. Figure 7.2 shows a representation of the package, the cells alone would add up to 126 kg. It is necessary to set with the manufacturer the total weight of the component, since it is missing considerations like refrigerant, regulator and wiring.

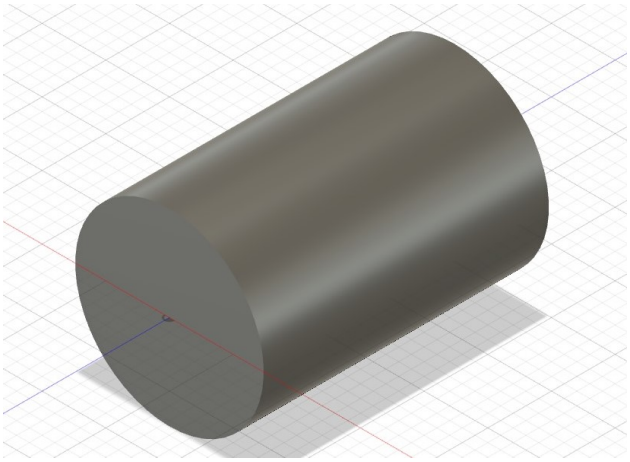


Figure 7.2: Battery package

8 Weight engineering

In the table 8.1 are presented the estimated weights of the elements that compose the aircraft. The fuselage nose was used as datum to estimate the relative positions and center of gravity (CG). The total estimated weight of the aircraft using the elements presented in the table is **2678 kg**.

Table 8.1: Weights

Element	Quantity	Weight per unit	Weight (kg)	Element position respect to datum (x axis) in mm	Element position respect to datum (y axis) in mm
Wing	1	54	54	3400	1400
Tilt motor	2	50	100	1730	2026
VTOL motor 1	2	50	100	1700	1700
VTOL motor 2	2	50	100	5100	2078
VTOL motor 3	2	50	100	5600	2078
Fuselage structure	1	358	358	3550	420
Long booms	2	26	54	6460	1375
Short booms	2	16	32	3691	1375
Pax. pos 1	2	88	176	2264	361
Pax. pos 2	2	88	176	3125	361
Pax. pos 3	2	88	176	4265	361

Baggage	1	30	30	5800	300
H. tail	1	16	16	7400	1735
V. tail	2	2	3	7400	1735
Nose landing gear	1	35	35	1410	-600
Main landing gear	2	52,5	105	4550	-600
Battery pack	8	126	1008	3500	1520
Avionics	1	30	30	600	200
Cockpit windshield	1	25	25	1857	986

9 Stability and control estimation analysis

xflr5 software was used to estimate the stability and control parameters of the aircraft. fig 9.1 shows that the aircraft is statically stable.Likewise, figure 9.2 and fig 9.3 show that the aircraft is dynamically stable both longitudinally and latitudinally.

On the other hand, according to the *thumbprint criterion*, the aircraft can be considered acceptable. The *thumbprint criterion* is a graphical tool used in aircraft design to assess the handling qualities of an aircraft. It plots the damping ratio (ζ) and undamped natural frequency (ω_n) of an aircraft's longitudinal dynamics to categorize the aircraft's response into qualitative levels of pilot acceptance, ranging from *Unacceptable* to *Satisfactory*. This criterion helps to ensure that the aircraft will behave in a way that is manageable and predictable for pilots.

According to the simulations and analysis conducted, the damping ratio (ζ) has a value of 0.375 and a natural frequency (ω_n) of 3.1 Hz.With these results, the aircraft is placed in the acceptable zone. fig 9.5 shows the exact location of the airplane (blue dot)

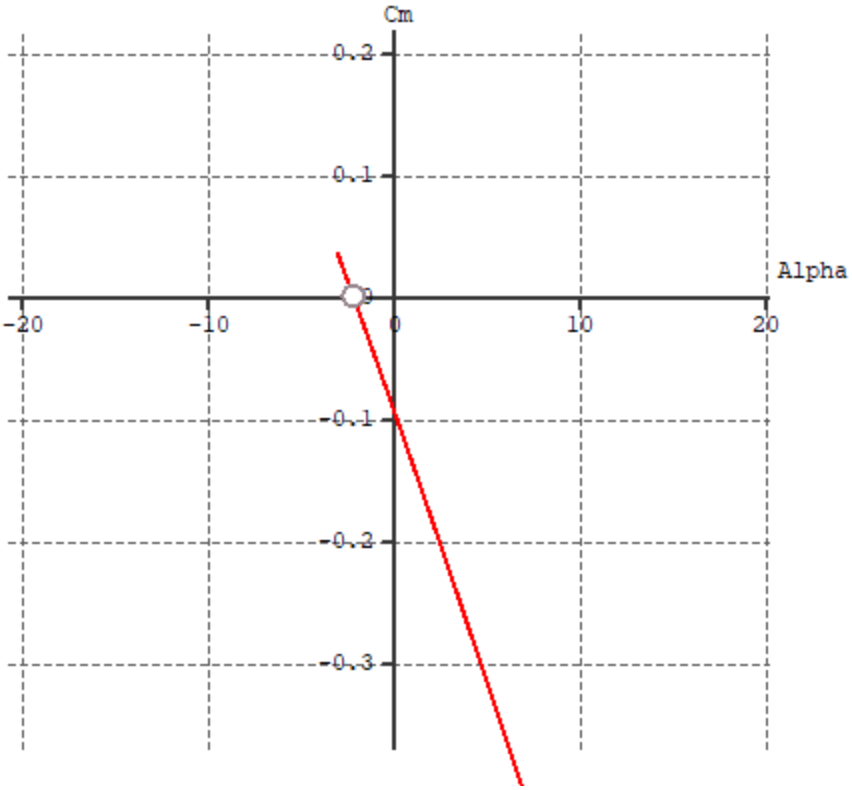


Figure 9.1: Static stability

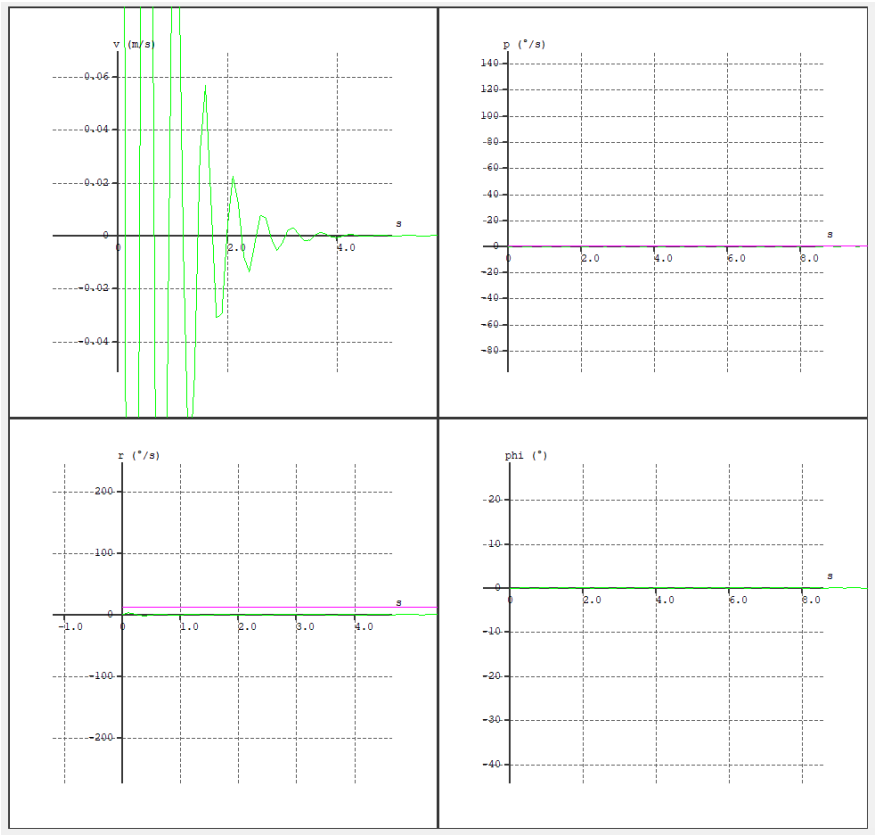


Figure 9.2: Lateral modes plot

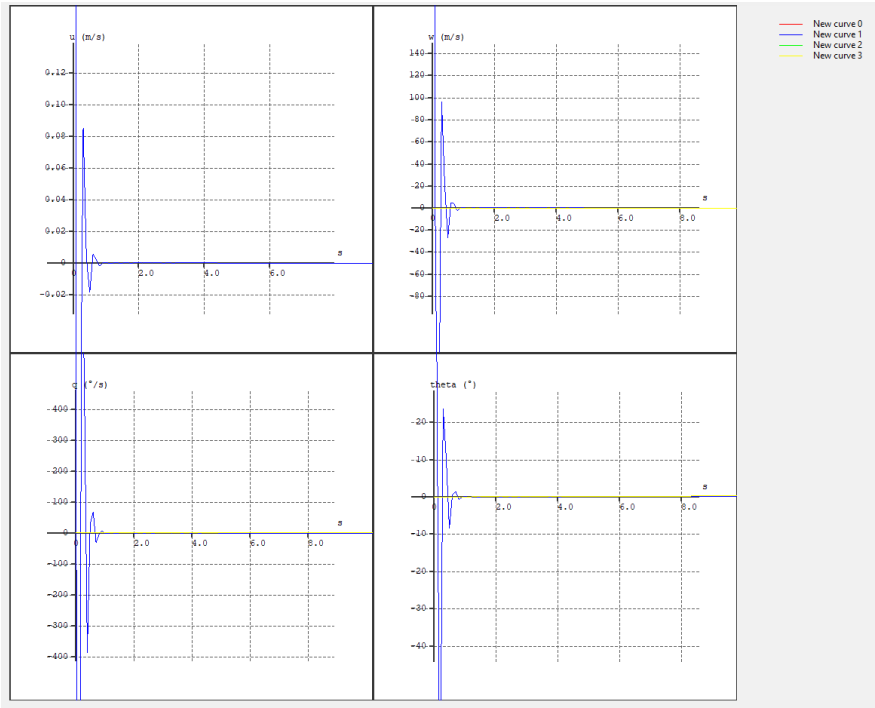


Figure 9.3: Longitudinal modes plot

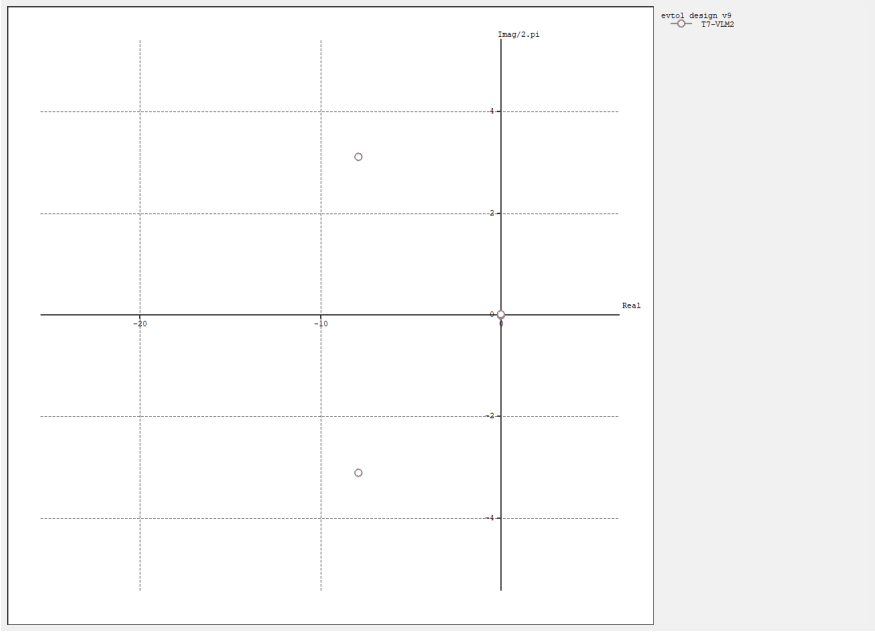


Figure 9.4: modes

If an aircraft is deemed *acceptable* according to the thumbprint criterion, it implies the following:

1. **Handling Qualities:** The aircraft's handling qualities in the longitudinal axis (pitch motion) are within a range that pilots generally consider acceptable. This means that the aircraft's response to control inputs in pitch, as well as its behavior following disturbances in pitch, is manageable and predictable to a satisfactory degree.

2. **Pilot Workload:** The pilot workload is likely reasonable during normal operations. The aircraft does not demand excessive attention or control input to maintain the desired flight path or attitude.

3. **Stability and Control:** The aircraft possesses a balance of damping and natural frequency that results in a stable platform without being overly damped (which could result in sluggish response) or under-damped (which could lead to oscillatory or uncomfortable behavior).

4. **Safety Margins:** Being within the *Acceptable* region typically indicates that there are safety margins in place. The aircraft should be able to cope with a reasonable range of operational conditions and disturbances without becoming unsafe.

5. **Design Adequacy:** For the particular phase of flight or flight condition that the graph represents, the design of the aircraft meets the basic criteria for stability and control effectiveness.

6. **Further Optimization Possible:** Although the aircraft is acceptable, there may be room for improvement. Engineers might aim to refine the design to push the performance into the *Satisfactory* region for an even better pilot experience and performance margin.

7. **Compliance with Standards:** The aircraft likely meets minimum standards set forth by regulatory bodies such as the Federal Aviation Administration (FAA) or the

European Union Aviation Safety Agency (EASA) for the particular aspect of handling being evaluated.

In summary, an aircraft meeting the *Acceptable* criteria on the thumbprint chart indicates it has passed a certain threshold in terms of its dynamic behavior, making it suitable for operation.

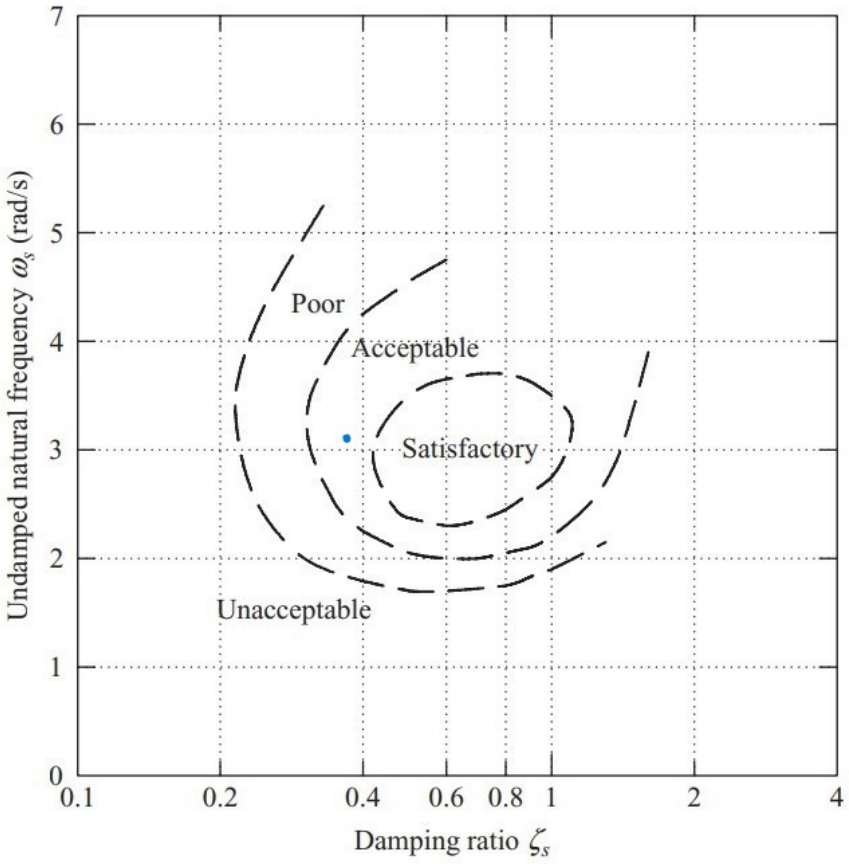


Figure 9.5: thumbprint

10 Conclusion

The revision of the first iteration of the design phase results in what was reported here. These results will drive the future work in the project with the following requirements:

- Stability and control derivatives revision
- Actual performance estimation
- Propulsion system integration (with motor and propeller manufacturer's collaboration).
- Energy system integration with the batteries form factor and actual capabilities.
- Producibility considerations and production design
- Conceptual design revision

These steps will bring the conceptual design of the aircraft to a converging stage, so the design can be considered completed and further steps can be made, considering all the factors necessary to launch a viable eVTOL aircraft to the market.

Bibliography

[1] S. Gudmundsson, "General aviation aircraft design - applied methods and procedures," *Butterworth-Heinemann*, 2022.

[2] Wikipedia, *List of STOL aircraft*. [Online]. Available: https://en.wikipedia.org/wiki/List_of_STOL_aircraft.

[3] *Emrax electric motors / generators*, https://emrax.com/e-motors/, Accessed: 2024-04-05, EMRAX d.o.o., 2024.

[4] *Integrated motor — controller: Magidrive™ - magicall*, https://www.magicall.biz/products/integrated-motor-controller-magidrive/, Accessed: 2024-04-05, MAGicALL, Inc.

[5] *Services - magnix*, https://www.magnix.aero/services, Accessed: 2024-04-05, magniX, 2024.

[6] OO, JH, and THO, "Siemens eaircraft - disrupting aircraft propulsion," Siemens AG, Technical Report, Apr. 2018, Accessed: 2024-04-05. [Online]. Available: https://www.ie-net.be/sites/default/files/Siemens%20eAircraft%20-%20Disrupting%20Aircraft%20Propulsion%20-%20OO%20JH%20THO%20-%2020180427.cleaned.pdf.

[7] *Bldc electric motors: 5–250 kw*, https://www.mgm-compro.com/electric-motors/, Accessed: 2024-04-05, MGM COMPRO, 2024.

[8] D. Raymer, *Aircraft design: a conceptual approach*. American Institute of Aeronautics and Astronautics, Inc., 2012.